Alex Stoken

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EDUCATION

University of Texas at Austin Austin, Texas
M.S. in Computer Science *May 2022*
Thesis: DexV2A: Vision Pretraining for Dexterous Manipulation, advised by Dr. Kristen Grauman

University of Arizona Honors College Tucson, Arizona
B.S. in Physics w/ Honors | B.S. in Mathematics | B.A. in Economics | Minors in Spanish, Astronomy *May 2019*
Thesis: Background Characterization in the 4Top Search at the ATLAS Experiment, advised by Dr. Erich Varnes

RESEARCH AND WORK EXPERIENCE

NASA's Astromaterials and Exploration Science Division, JETS contract Houston, TX
Data Scientist *June 2019 – Present*

Artemis Mission Operations

- Software lead for Crew Lunar Observations team, designing tools & ops flows for photography during lunar missions
- Developed software for Artemis Science mission backroom operations, in-flight crew use, and data archiving
- Built data visualization used in Mission Control front room during Artemis II lunar flyby
- Trained for and supported Imagery Data console operations in Imagery Ops team during the Artemis I & II missions

Geospatial Machine Learning & Analysis

- Build, train and deploy machine learning models to (1) detect surface features in remotely sensed lunar imagery, (2) estimate roughness where elevation model data is unavailable
- Developed software tools to (1) automate hazard analysis in lunar imagery and (2) improve efficiency in geospatial analysis
- Identify craters from digital elevation models, render lunar imagery, and train crater detection models for crater-based optical navigation project (TRON)

Crew Earth Observations (International Space Station payload)

- Created novel technique to localize & georeference day and nighttime astronaut photography (800k/5M photos localized)
- Build, train and deploy machine learning models to (1) enhance Gateway to Astronaut Photography of Earth database with geographic metadata, (2) identify and track ISS objects of interest
- Built interactive website to explore localized astronaut photos (`https://eol.jsc.nasa.gov/ExplorePhotos/`)
- Developed software to automate operational workflows for daily ISS payload operations and image analysis for ISS surveys
- Wrote 28 "Image of the Day" articles on astronaut photographs for NASA Earth Observatory
- Contribute to concept development and proposal writing for competitive funding calls
- Represent group at internal & external meetings, including ISS Imagery Working Group, POIWG, ISS R&D

NASA's Technology and Innovation Data Analytics Team Washington, D.C.
Data Science Intern *June 2018 – August 2018*

- Designed and built usable 3D immersive network visualization tool as lone developer (C#/Unity).
- Document findings, lessons learned, and final codebase; showcase with Chief Information Officer Senior Staff.

NASA/Arizona Space Grant Consortium Tucson, AZ
Student Researcher in the Dept. of Physics *August 2017 – May 2018*

- Improved vector-like quark detection by 10% over traditional methods using machine learning and Bayesian statistics in Python with scikit-learn and CERN's ROOT package.
- Performed feature selection, hyperparameter optimization, and algorithm comparison to find the best model.

US Dept. of Energy - Italian Natl Institute for Nuclear Physics Univ. of Bari Aldo Moro, Italy
High Energy Physics Summer Student Researcher *June – July 2017*

- Applied multivariate analysis techniques (python) to test dark matter models for consistency with current physics.
- Collaborated with international partners on formal written summary of findings for CERN internal publication.

University of Arizona Dept. of Systems Engineering Tucson, AZ
IBM Watson Sports Analyst *January 2016 – May 2016*

- Examined MLB data with IBM Watson to find trends between player performance and game scheduling.
- Developed tutorial for students to learn the software; completed Watson Fundamentals course on Big Data.

University of Arizona Dept. of Mathematics Tucson, AZ
Student Researcher *January 2016 – May 2016*

- Analyzed sparse collegiate golf data with R to find trends for individual UArizona golf players.
- Presented individual and team recommendations to head golf coach.

TEACHING AND SERVICE

Teaching Assistant | University of Texas at Austin Spring 2022, Spring 2025
Online Learning and Optimization (Graduate Course) Advances in Deep Learning

- Hosted office hours and exam review sessions, answered student content and logistics questions.

Preceptor | University of Arizona Spring 2017 – Spring 2019
Physics I, Physics II

- Mentored 150 students per semester in building physical intuition and developing problem solving skills.

Professional Service

- Reviewer: CVPR ('24, '26), ECCV ('26), ACMM ('25), NeurIPS ('26),NASA SBIR Proposals ('24, '26)
- Intern Mentor (3x)
- Scholarship Interview Committee ('26)

Community Service

- STEM Judge: Houston Science Fair (3x), FIRST Lego League (2x), Tucson Science Fair (2x), Conroe Challenge (1x)
- Letters to a Pre-Scientist Pen Pal (2x)
- Open-Source Contributions: vismatch (maintainer), Keras-SWA, Ultalytics YOLO

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

JETS Educational Outreach Group
Leadership Team *2020 – Present*

- Coordinating STEM outreach activities between local schools and JETS Contract workforce.
- Volunteering at multiple activities per year with local middle and elementary schools.
- Build team for AR sandbox education tool.

Johnson Space Center (JSC) Hackathon
Co-Director *2021 – 2023*

- Redesigned hackathon as an external facing event attended by over 300 people, including community partners, students, and members of the JSC workforce.
- Hosted (1) workshop on computer vision at NASA and (2) tutorial on how to get started in deep learning.
- Coordinated talks from NASA subject matter experts and developed challenges focused on JSC & Artemis missions.

Johnson Space Center EMERGE Early Career Resource Group
Leadership Team *2020 – 2022*

- Initiated Lightning Talk series to develop member's public speaking skills and foster community engagement.
- Organized and hosted bimonthly personal and professional development activities and lectures.

Arizona Model United Nations Tucson, AZ
President | Director of Internal Affairs *Aug 2016 – May 2019*

- Remodeled fundraising strategy to reduce membership costs by 35% for 75 students.
- Selected by U.S. Consulate in Hermosillo, MX to speak to 300 high school students on leadership and diplomacy.
- Led recruitment and retention efforts which resulted in a 18% increase in club membership.
- Best Delegate Award at American Model United Nations 2016 Conference (representing Kazakhstan).

HackArizona Tucson, AZ
Multiple Projects *Jan 2017/2018/2019*

- **Project: HERE (1st place Overall Hack):** Acted as project lead, presenter, and UX designer for Android location-based driver assist app.
- **Project: Disaster Relief (1st place Best use of VR/AR to Help Others):** Built and presented Android AR app to help Emergency Medical Services locate and prioritize victim care at disaster sites.
- **Project: Neural Network Visualizer (2nd Place in Best Use of Data):** Built and presented Unity VR visualization showcasing differences between convolutional neural network and human image understanding.

Camp Wildcat Tucson, AZ
Auction Director *Oct 2016 – April 2018*

- Co-chaired annual fundraising auction for more than 80 guests; raised $6,500 for local Title 1 students to go on empowerment and self-betterment camping trips.

Other Activities and Memberships: IEEE/CVF Member, Mortarboard Honor Society, Arizona Sports Analysis Assoc.

PUBLICATIONS AND PRESENTATIONS

Conference Papers (Peer Reviewed)

- [**CVPRW 2026**] I Corley, **A Stoken**, G Berton. "Are Pretrained Image Matchers Good Enough for SAR-Optical Satellite Registration?". *IEEE/CVPR Image Matching Workshop*, 2026.
- [**WACV 2026**] P Mandikal, T Nagarajan, **A Stoken**, Z Xue, K Grauman. "SPOC: Spatially-Progressing Object State Change Segmentation in Video". *WACV*, 2026.
- [**ICCV 2025**] G Berton, **A Stoken**, C Masone. "AstroLoc: Robust Space to Ground Image Localizer". *ICCV*, 2025.
- **[CVPRW 2024] A Stoken**, P Ilhardt, M Lambert, K Fisher. "(Street) Lights Will Guide You: Georeferencing Nighttime Astronaut Photography of Earth". *IEEE/CVPR EarthVision Workshop*, 2024.
- **[CVPRW 2023] A Stoken**, K Fisher. "Find My Astronaut Photo: Automated Localization and Georectification of Astronaut Photography". *IEEE/CVPR Image Matching Workshop*, 2023.
- [**CVPR 2024**] G Berton, **A Stoken**, B Caputo, C Masone. "EarthLoc: Astronaut Photography Localization by Indexing Earth from Space". *IEEE/Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [**CVPRW 2024**] G Berton, G Goletto, G Trivigno, **A Stoken**, B Caputo, C Masone. "EarthMatch: Iterative Coregistration for Fine-Grained Localization of Astronaut Photography". *IEEE/Computer Vision and Pattern Recognition Image Matching Workshop (CVPRW)*, 2024.

Conference Abstracts (Peer Reviewed)

- **A Stoken**, P Ilhardt, S Walton, H Danuqe, M Thomas, P Montalvo, C Pittman, A Britton. "Robust, High Precision Automated Coregistration of LROC NAC Imagery". *Lunar and Planetary Science Conference*, 2025.
- S Schmidt, **A Stoken**, W Garcia-Lopez, C Boyer, D Charney, M Miller. "Enabling Orbital Crew Lunar Observations During Artemis Missions". *Lunar and Planetary Science Conference*, 2025.
- S Schmidt, **A Stoken**, W Garcia-Lopez, C Boyer, D Charney, M Miller. "Artemis II Lunar Observations Support Software". *Lunar and Planetary Science Conference*, 2026.
- **A Stoken**, P Ilhardt, A Britton. "Learning Terrain Ruggedness from LROC NAC Image Data". *Lunar and Planetary Science Conference*, 2024.
- **A Stoken**, A Britton, M Lambert, A Turner, M Rubio. "Automated Boulder Counting: Deep Learning for Boulder Detection and Height Estimation". *Lunar and Planetary Science Conference*, 2023.
- C Lawson, P Ilhardt, **A Stoken**, M Evans. "Surface Gravimetry Using Rover Navigation Systems". *Lunar and Planetary Science Conference*, 2024.
- M Rubio, P Ilhardt, **A Stoken**, S Walton. "Characterizing Small Craters in the Lunar South Pole Using the Crater Morphology Profile Tool (CAMEO)". *Lunar and Planetary Science Conference*, 2024.

Internal Publications (Peer Reviewed)

- N De Filippis, G Miniello, D Burns, M Mulhearn, H Prosper, S Tentindo, R Mohamed Aly, S Elgammal, **A Stoken**, [3 others] (2016) "Search for Dark Matter Produced in Association with a Higgs Boson in the four lepton final state at 13 TeV" CMS Analysis EXO-18-009.

Presentations

- "Improving Spaceflight Imagery with Machine Learning" - Intl. Workshop on AI Powered Space ('23)
- "CEO: What's new in Astronaut Photography" - Payload Ops and Integ. Working Group Mtg #52 ('23)
- Panelist, "AI at NASA Centers" - 3rd NASA AI and Data Science Workshop ('23)
- "Find My Astronaut Photo: Automated Localization of Imagery" - ISS R&D Conference ('22)
- "Understanding ISS Imagery through the Lens of Machine Learning" - Jacobs Sci Dept. Seminar Series ('20)

- "Efficacy of Super-Modeling in Climate Systems" - UArizona Math Department Poster Session ('19)
- "Machine Learning Applications in HEP: Search for Vector-like Quarks" - NASA/Arizona Space Grant Symposium ('18)
- "Reading Between the Strokes: Collegiate Golf Analytics" - UArizona Honors Engagement Expo ('16)

HONORS AND AWARDS

Professional

- JSC Director's Commendation (2x) | EAISD Directorate, NASA Johnson Space Center
- Annual Award for Continuous Improvement ('23) | JETS Contract
- NASA On the Spot - NASA Johnson Space Center
- Quarterly Individual Award ('22) | JETS Contract
- Quarterly Team Award ('20, '21, '22, '23) | JETS Contract
- ROCS Award ('19, '22, '23, '24) | JETS Science and Exploration Department
- TRON the Spot Award ('26) — TRON ECI project
- Group Special Act Award ('26) — NASA Johnson Space Center - ARES Artemis II Lunar Science Team
- Group Special Act Award ('26) — NASA Johnson Space Center - AMPES Artemis Landing Mission Support
- JSC Group Achievement Award ('26) - Artemis II Lunar Science Flight Control Team

Scholastic

- **National:** United States Presidential Scholar ('15); Coca-Cola Scholar ('15)
- **State:** Flinn Scholar ('15) (Top 20 AZ seniors, $150k scholarship)
- **Institutional:** Galileo Circle Scholar ('19); Weaver Award ('18); Purviance Award ('16); Dr. McBreaty Scholarship ('16)

GRANTS

- **Internal R & D Grant (2026):** NASA JSC - $100k to train an image matcher for space applications.
- **Internal R & D Grant (2021-24):** NASA JSC - $100k/year to train ML models to ID spaceflight imagery.
- **SABRE Grant (2026):** JETS Contract - $10k to build deep learning model to directly build DEMs from NAC imagery
- **SABRE Grant (2022):** JETS Contract - $10k to build deep learning model to id boulders in the lunar South Pole.
- **SABRE Grant (2021):** JETS Contract (Project Lead) - $10k to prototype water segmentation via satellite imagery.
- **Internal R & D Grant (2021):** NASA JSC - $100k to identify astronaut pose during intravehicular activities.
- **NASA/Arizona Space Grant (2018):** Arizona Space Grant Consortium.
- **Professional Development Grant (2018):** Honors College.
- **Travel Grant (2017):** Honors College.
- **Spirit of Inquiry Research Grant (2016):** Honors College.

TECHNICAL SKILLS

Python Proficiency

ML: PyTorch, TF/Keras, Transformers, Lightning
Imagery: OpenCV, skimage, PIL, COLMAP
Data Analysis: numpy, pandas, scipy
Geospatial: rasterio, geopandas
Visualization: matplotlib, pyvista, rerun
GUI: pyQT, streamlit, Gradio

Programming

C#
MATLAB
TS/JS
Java
LaTeX
Bash

Technical Programs

ArcGIS
Git
Microsoft Office Suite
Photoshop
Unity
STK

Language

English (native)
Spanish (conversational)
Italian (beginner)